

Emulacija računalnika ZX Spectrum z vpogledom v stanje sistema

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Povzetek

- ZX Spectrum 48K: zgodovina, zgradba
- cilj dela
- razvoj in končna implementacija
- pregled in ovrednotenje končnih rezultatov

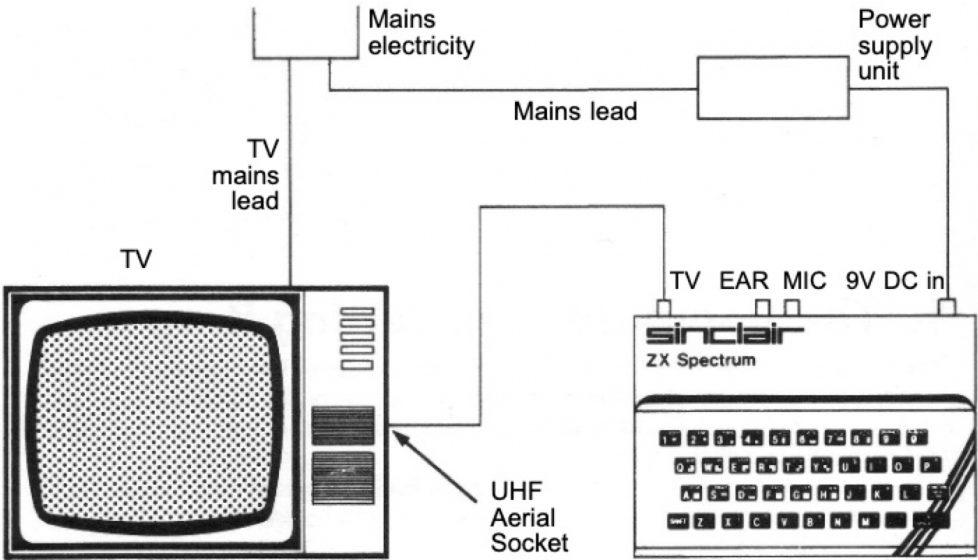
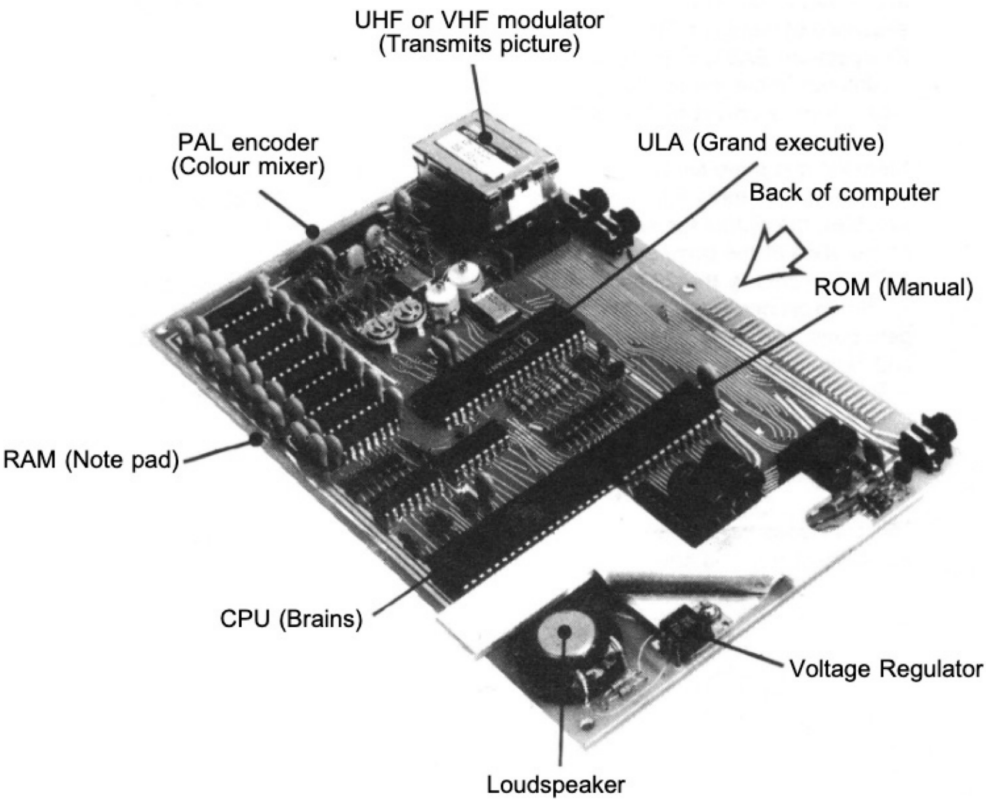


ZX Spectrum 48K

Zgodovina



Zgradba



Cilj dela

Cilji dela

Registers

Main register set

A	FA	F	32				
B	C6	C	05				
D	1F	E	26				
H	59	L	FA				
S	Z	F5	H	F3	PV	N	C
0	0	1	1	0	0	1	0

Alternate register set

A'	00	F'	44				
B'	10	C'	00				
D'	36	E'	9B				
H'	27	L'	58				
S	Z	F5	H	F3	PV	N	C
0	1	0	0	0	1	0	0

Special purpose registers

I	3F	IX	0000
R	5C	IY	84A1
		SP	9CFC
		PC	9302

Disassembly

Spare space

9302	1D	DEC E
9303	20 05	JR NZ, +05h
9305	FD 5E 02	LD E, (IY+02h)
9308	EE 18	XOR 18h
930A	10 EC	DJNZ -14h
930C	0D	DEC C
930D	20 50	JR NZ, -17h

Memory

Spare space

9300	EE 18 1D 20 05 FD 5E 02 EE 18 10 EC 0D 20 E9 CD
9310	37 93 C0 FD 7E 01 CD 2B 93 36 38 FD 7E 02 CD 2B
9320	93 36 38 FD 23 FD 23 FD 23 18 B1 D6 08 0F 0F 0F
9330	2F F6 E0 6F 26 59 C9 3A 59 84 B7 28 05 DB 1F CB
9340	67 C0 01 FE BF ED 78 E6 01 FE 01 C9 09 44 45 43
9350	09 45 78 0F 0D 09 4A 52 09 4E 5A 2C 4E 4F 46 4C
9360	50 36 82 0F 0A 09 4C 44 09 45 2C 28 48 4C 29 8C
9370	0F 07 09 58 4F 52 09 32 34 96 0F 10 4E 4F 46 4C
9380	50 36 09 44 4A 4E 5A 09 54 4D 35 31 A0 0F 06 09
9390	44 45 43 09 43 AA 0F 0B 09 4A 52 09 4E 5A 2C 54
93A0	4D 35 31 B4 0F 13 4E 4F 4E 4F 54 45 34 09 4C 44

subpectrum screen



9330

2F

CPL

9331

FA F0

OR F0h

Razvoj

Moduli funkcionalnosti:

- Processor
- Memory
- ULAScreen , ULAKeyboard ,
ULATapeDeck , ULABeeper

The screenshot displays the 'subspectrum' emulator interface, which is divided into several panels:

- Registers:** This panel shows the state of the processor's registers. It includes a 'Main register set' (A, B, C, D, E, F, H, L, S, Z, F5, H, F3, PV, N, C), an 'Alternate register set' (A', B', C', D', E', F', H', L', S, Z, F5, H, F3, PV, N, C), and 'Special purpose registers' (I, R, IX, IY, SP, PC).
- Disassembly:** This panel shows the current instruction being executed, along with its operands and the state of the 'Spare space' (memory).
- Memory:** This panel shows the current memory address and its contents, along with the state of the 'Spare space' (memory).
- Game Screen:** A small window titled 'subspectrum screen' shows a game titled 'MANIC MINER' with the text 'PRESS ENTER TO START' and a small character.

Pomnilnik

Memory



15E0	15	D8	28	FA	CF	07	D9	E5	2A	51	5C	23	23	18	08	1E
15F0	30	83	D9	E5	2A	51	5C	5E	23	56	EB	CD	2C	16	E1	D9
1600	C9	87	C6	16	6F	26	5C	5E	23	56	7A	B3	20	02	CF	17
1610	1B	2A	4F	5C	19	22	51	5C	FD	CB	30	A6	23	23	23	23
1620	4E	21	2D	16	CD	DC	16	D0	16	00	5E	19	E9	4B	06	53
1630	12	50	1B	00	FD	CB	02	C6	FD	CB	01	AE	FD	CB	30	E6
1640	18	04	FD	CB	02	86	FD	CB	01	8E	C3	4D	0D	FD	CB	01
1650	CE	C9	01	01	00	E5	CD	05	1F	E1	CD	64	16	2A	65	5C
1660	EB	ED	B8	C9	F5	E5	21	4B	5C	3E	0E	5E	23	56	E3	A7
1670	ED	52	19	E3	30	09	D5	EB	09	EB	72	2B	73	23	D1	23
1680	3D	20	E8	EB	D1	F1	A7	ED	52	44	4D	03	19	EB	C9	00
1690	00	EB	11	8F	16	7E	E6	C0	20	F7	56	23	5E	C9	2A	63
16A0	5C	2B	CD	55	16	23	23	C1	ED	43	61	5C	C1	EB	23	C9
16B0	2A	59	5C	36	0D	22	5B	5C	23	36	80	23	22	61	5C	2A
16C0	61	5C	22	63	5C	2A	63	5C	22	65	5C	E5	21	92	5C	22
16D0	68	5C	E1	C9	ED	5B	59	5C	C3	E5	19	23	7E	A7	C8	B9
16E0	23	20	F8	37	C9	CD	1E	17	CD	01	17	01	00	00	11	E2
16F0	A3	EB	19	38	07	01	D4	15	09	4E	23	46	EB	71	23	70
1700	C9	E5	2A	4F	5C	09	23	23	23	4E	EB	21	16	17	CD	DC
1710	16	4E	06	00	09	E9	4B	05	53	03	50	01	E1	C9	CD	94
1720	1E	FE	10	38	02	CF	17	C6	03	07	21	10	5C	4F	06	00
1730	09	4E	23	46	2B	C9	EF	01	38	CD	1E	17	78	B1	28	16
1740	EB	2A	4F	5C	09	23	23	23	7E	EB	FE	4B	28	08	FE	53
1750	00	04	FF	5C	00	0F	00	FD	47	70	00	70	00	FF	00	54

Memory



ROM

THE 'SAVE, LOAD, VERIFY and MERGE' COMMAND ROUTINES

15D4 FD CB 02 0E 20 04 FD CB 02 0E CB E0

15E0 15 D8 28 FA CF 07

THE 'INPUT-AD' SUBROUTINE

15E6 D9 E5 2A 51 5C 23 23 18 08

THE 'MAIN PRINTING' SUBROUTINE

This subroutine is called with the A register holding a valid stream number normally +FD to +03. Then depending on the stream data a particular channel will be made the current channel.

56 EB CD 2C 16 E1 D9

THE 'CHAN-OPEN' SUBROUTINE

1601 87 C6 16 6F 26 5C 5E 23 56 7A B3 20 02 CF 17

1610 1B 2A 4F 5C 19

THE 'CHAN-FLAG' SUBROUTINE

1615 22 51 5C FD CB 30 A6 23 23 23 23

1620 4E 21 2D 16 CD DC 16 D0 16 00 5E 19 E9

THE 'CHANNEL CODE LOOK-UP' TABLE

162D 4B 06 53

1630 12 50 1B 00

THE 'CHANNEL 'K' FLAG' SUBROUTINE

1634 FD CB 02 C6 FD CB 01 AE FD CB 30 E6

1640 18 04

THE 'CHANNEL 'S' FLAG' SUBROUTINE

1642 FD CB 02 86 FD CB 01 8E C3 4D 0D

THE 'CHANNEL 'L' FLAG' SUBROUTINE

Centralna procesna enota Z80

Registri in zastavice

Registers

Main register set								Alternate register set							
A	00			F	5C			A'	00			F'	44		
B	17			C	4B			B'	00			C'	00		
D	00			E	06			D'	5C			E'	B9		
H	10			L	7F			H'	00			L'	00		
S	Z	F5	H	F3	PV	N	C	S	Z	F5	H	F3	PV	N	C
0	1	0	1	1	1	0	0	0	1	0	0	0	1	0	0

Special purpose registers							
I	3F			IX	0000		
R	31			IY	5C3A		
				SP	FF4E		
				PC	15E1		

Ukazi

CPE Z80:

- 158 ukazov
- 10 tipov naslavljanja
- 3 načini delovanja prekinitev

00rrrr110 nnnnnnnn → LD r,n

Disassembly



ROM

THE 'WAIT-KEY' SUBROUTINE

○ 15D8 20 04 1B N7 +04h
○ 15DA FD
Call the input subroutine indirectly via INPUT_AD.

○ 15DE CD E6 15 CALL 15E6h

○ 15E1 D8 RET C

○ 15E2 28 FA JR Z, -06h

○ 15E4 CF RST 08h

○ 15E5 07 RLCA

THE 'INPUT-AD' SUBROUTINE

○ 15E6 D9 EXX'

○ 15E7 E5 PUSH HL

○ 15E8 2A 51 5C LD HL, (5C51h)

○ 15EB 23 INC HL

○ 15EC 23 INC HL

○ 15ED 18 08 JR +08h

THE 'MAIN PRINTING' SUBROUTINE

○ 15EF 1E 30 LD E, 30h

○ 15F1 83 ADD A, E

○ 15F2 D9 EXX'

○ 15F3 E5 PUSH HL

○ 15F4 2A 51 5C LD HL, (5C51h)

○ 15F7 5E LD E, (HL)

○ 15F8 23 INC HL

○ 15F9 56 LD D, (HL)

○ 15FA EB EX DE, HL

Izvajanje

Processor.step :

- prebere ukaz
- posodobi programski števec
- ukaz izvede
- poveča števec izvedenih ukazov
- po potrebi proži prekinitve

Disassembly



ROM

THE 'WAIT-KEY' SUBROUTINE

```
O 15D8  20 04 1B N7 +04h  
O 15DA  FD  
O 15DE  CD E6 15  CALL 15E6h  
O 15E1  D8  RET C  
O 15E2  28 FA  JR Z, -06h  
O 15E4  CF  RST 08h  
O 15E5  07  RLCA
```

Call the input subroutine indirectly via INPUT_AD.

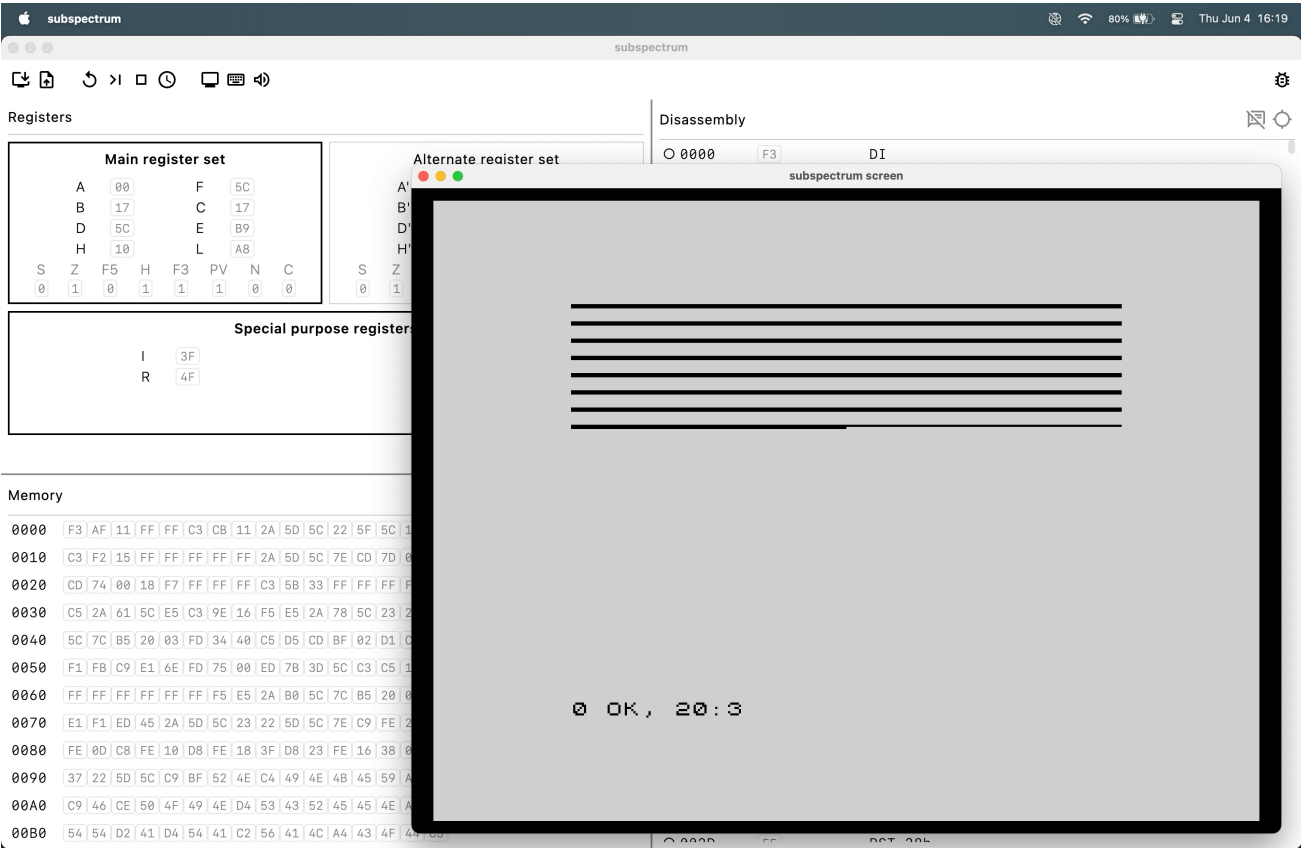
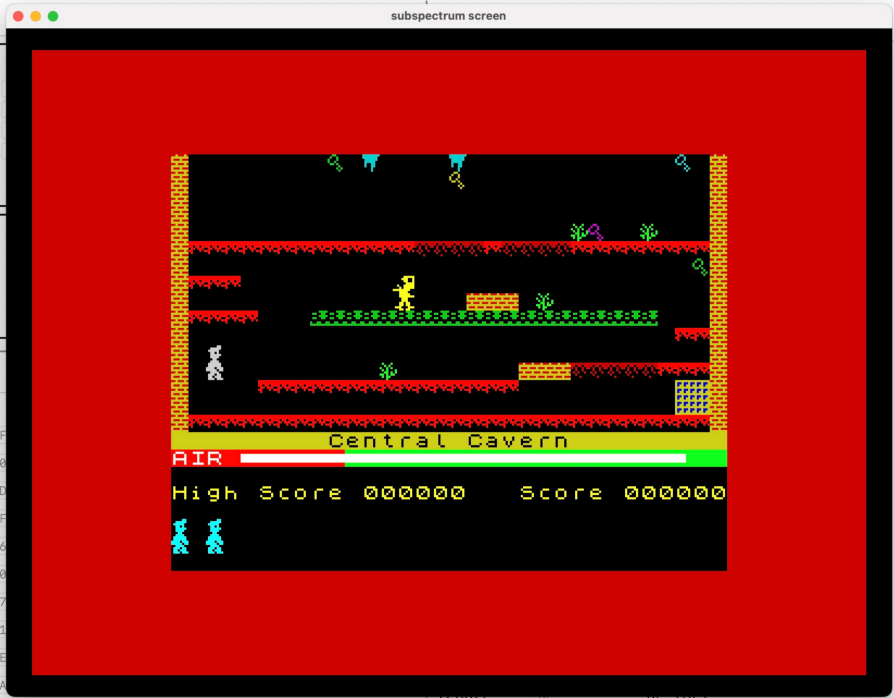
THE 'INPUT-AD' SUBROUTINE

```
O 15E6  D9  EXX'  
O 15E7  E5  PUSH HL  
O 15E8  2A 51 5C  LD HL, (5C51h)  
O 15EB  23  INC HL  
O 15EC  23  INC HL  
O 15ED  18 08  JR +08h
```

THE 'MAIN PRINTING' SUBROUTINE

```
O 15EF  1E 30  LD E, 30h  
O 15F1  83  ADD A, E  
O 15F2  D9  EXX'  
O 15F3  E5  PUSH HL  
O 15F4  2A 51 5C  LD HL, (5C51h)  
O 15F7  5E  LD E, (HL)  
O 15F8  23  INC HL  
O 15F9  56  LD D, (HL)  
O 15FA  EB  EX DE, HL
```

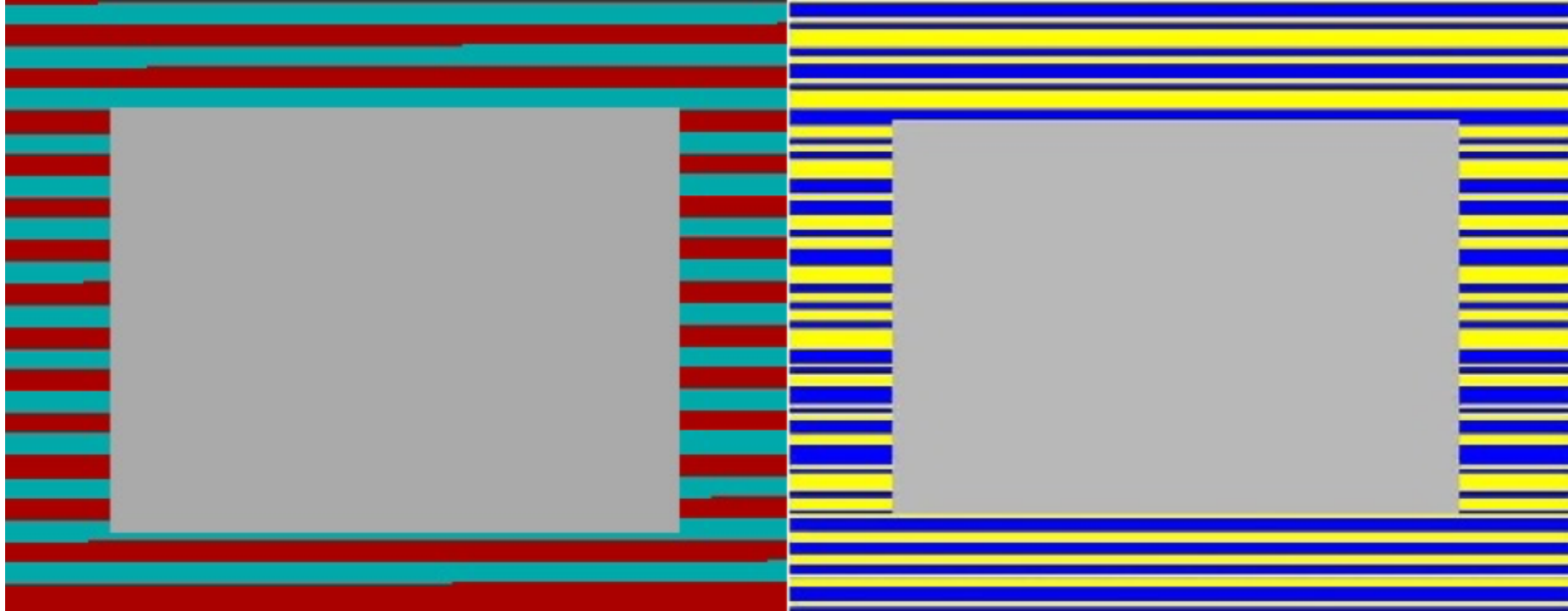
ULA



Tipkovnica



Kasete



Zvok



Evalvacija

Pravilnost emulacije - Manic Miner

sub spectrum

sub spectrum

Registers

Main register set

A	00	F	44
B	0A	C	C9
D	45	E	37
H	65	L	37
S	0	Z	1
F5	0	H	0
F3	0	PV	1
N	0	C	0

Alternate register set

A'	00	F'	44
B'	10	C'	00
D'	36	E'	9B
H'	27	L'	58
S	0	Z	1
F5	0	H	0
F3	0	PV	1
N	0	C	0

Special purpose registers

I	3F	IX	83D0
R	28	IY	808E
		SP	9CFE
		PC	87AB

Disassembly

87A0	ED B0	LDIR
87AD	3A 58 84	LD A, (8458h)
87B0	B7	OR A
87B1	28 15	JR Z, +15h
87B3	3D	DEC A

Memory

System variables

8790	07 84 FE 0B CC 35 91 3A 07 84 FE 12 CC 73 8D CD
87A0	C5 8F 21 00 60 11 00 40 01 00 10 ED B0 3A 58 84
87B0	B7 28 15 3D 32 58 84 07 07 07 E6 38 21 00 5C 11
87C0	01 5C 01 FF 01 77 ED B0 21 00 5C 11 00 58 01 00
87D0	02 ED B0 DD 21 29 84 11 7A 50 0E 06 CD BA 92 DD
87E0	21 1F 84 11 6B 50 0E 06 CD BA 92 CD 3C 8A CA FF
87F0	88 01 FE FE ED 78 5F 06 7F ED 78 B3 E6 01 CA CC
8800	85 06 FD ED 78 E6 1F FE 1F 28 0A 06 02 ED 78 E6
8810	1F FE 1F 28 F6 3A 6B 80 FE FF CA FF 88 06 BF 21
8820	5C 84 ED 78 E6 1F FE 1F 28 0A CB 46 20 08 7E EE
8830	03 77 18 02 CB 86 CB 4E 20 25 3A 5B 84 3C 32 5B

sub spectrum screen

Central Cavern

AIR

High Score 000100 Score 000200

Pravilnost emulacije - INES

subpectrum

subpectrum

Registers

Main register set

A

27

B

FE

D

FF

H

00

S

0

Z

0

F5

0

H

0

F3

0

PV

0

N

0

C

1

F

0A

C

FE

E

27

L

2D

Alternate register set

A'

00

B'

17

D'

36

H'

27

S

0

Z

1

F5

0

H

0

F3

0

PV

1

N

0

C

0

F'

44

C'

21

E'

9B

L'

58

Special purpose registers

I

3F

R

57

IX

0000

IY

5C3A

SP

6D2D

PC

F291

Memory

Machine stack

F290

FE

FE

FF

28

EC

32

D2

FE

3E

FF

32

D3

FE

3E

00

32

F2A0

D4

FE

18

DD

1D

43

14

20

1E

7B

FE

18

28

E0

FE

27

F2B0

28

DC

0C

0D

20

63

21

05

02

06

00

4B

09

7E

FE

41

F2C0

38

02

C6

20

4F

18

57

15

7A

FE

27

20

42

7B

FE

18

F2D0

20

04

0E

01

18

AB

21

05

02

06

00

4B

09

7E

4F

FE

F2E0

30

38

3B

FE

3A

30

37

FE

32

20

04

0E

A0

18

2F

FE

F2F0

33

20

04

0E

A1

18

27

FE

34

20

04

0E

A4

18

1F

FE

F300

39

20

04

0E

9F

18

17

D6

30

5F

21

60

02

18

03

21

F310

30

FF

4B

06

00

09

4E

18

05

21

08

FF

18

F4

43

3A

F320

D3

FE

BB

28

07

3E

00

32

D4

FE

18

30

CD

CB

F3

3A

F330

D4

FE

16

00

3C

FE

01

20

04

1E

23

18

10

FE

07

30

Disassembly

F2A4

1D

F2A5

43

F2A6

14

F2A7

20

1E

F2A9

7B

F2AA

FE

18

DEC E

LD B, E

INC D

JR NZ, +1Eh

LD A, E

CP 18h

subpectrum screen

subpectrum je nov emulator sistema ZX Spectrum+

INES je urejevalnik besedit, ki znotraj subpectruma deluje su+

per



sub spectrum

sub spectrum

Registers

Main register set

A	00	F	44				
B	0A	C	C9				
D	45	E	37				
H	65	L	37				
S	Z	F5	H	F3	PV	N	C
0	1	0	0	0	1	0	0

Alternate register set

A'	00	F'	44				
B'	10	C'	00				
D'	36	E'	9B				
H'	27	L'	58				
S	Z	F5	H	F3	PV	N	C
0	1	0	0	0	1	0	0

Special purpose registers

I	3F	IX	83D0
R	28	IY	808E
		SP	9CFE
		PC	87AB

Disassembly

087A0	01 00 10	LD B0, 1000h
087AB	ED B0	LDIR
087AD	3A 58 84	LD A, (8458h)
087B0	B7	OR A
087B1	28 15	JR Z, +15h
087B3	3D	DEC A

Memory

System variables

8790	07 84 FE 0B CC 35 91 3A 07 84 FE 12 CC 73 8D CD
87A0	C5 8F 21 00 60 11 00 40 01 00 10 ED B0 3A 58 84
87B0	87 28 15 3D 32 58 84 07 07 07 E6 38 21 00 5C 11
87C0	01 5C 01 FF 01 77 ED B0 21 00 5C 11 00 58 01 00
87D0	02 ED B0 DD 21 29 84 11 7A 50 0E 06 CD BA 92 DD
87E0	21 1F 84 11 68 50 0E 06 CD BA 92 CD 3C 8A CA FF
87F0	88 01 FE FE ED 78 5F 06 7F ED 78 B3 E6 01 CA CC
8800	85 06 FD ED 78 E6 1F FE 1F 28 0A 06 02 ED 78 E6
8810	1F FE 1F 28 F6 3A 6B 80 FE FF CA FF 88 06 BF 21
8820	5C 84 ED 78 E6 1F FE 1F 28 0A CB 46 20 08 7E EE
8830	03 77 18 02 CB 86 CB 4E 20 25 3A 58 84 3C 32 5B

sub spectrum screen